# ABHINAV GOEL

**ROBOTICS ENGINEER | AUTONOMOUS SYSTEMS | UAV SPECIALIST**

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⌥ GitHub ▶ YouTube


## PROFESSIONAL SUMMARY

Robotics Engineer specializing in **ROS2 autonomous navigation, SLAM, sensor fusion, and vision-guided UAV systems**. Proven track record of designing, simulating, and hardware-testing autonomous mobile robots (AMRs) and multirotor drones from mathematical modeling and Gazebo SITL to field deployment. **Top 16 / 400+ teams (Top 4% nationally)** at IIT Bombay e-Yantra 2025–26. Skilled in ROS2 Humble, ArduPilot, PX4, MAVSDK, Pixhawk controllers, OpenCV, YOLO, and embedded hardware development.

## TECHNICAL SKILLS

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|---|---|
| **Autonomous & Controls:** | Autonomous Systems, PID Control, Sensor Fusion, EKF (Extended Kalman Filter), RANSAC, ROS2 Control, Visual Servoing, Trajectory Planning, Kinematics |
| **UAV & Multirotors:** | MultiRotor UAVs, Pixhawk Flight Controller, ArduPilot, PX4 Autopilot, QGroundControl, Mission Planner, MAVLink, MAVSDK, DroneKit |
| **Robotics & SLAM:** | ROS2 (Humble/Foxy), Nav2 Stack, MoveIt2, SLAM Toolbox, ORB-SLAM3, AMCL, TF2 Transformation Tree, URDF / xacro |
| **Perception & Vision:** | OpenCV, YOLOv8 / YOLOv4-Tiny, ArUco Marker Geolocation, Pinhole Camera Model |
| **Languages & Simulation:** | C++, Python, Gazebo, RViz2, ArduPilot SITL, MATLAB / Simulink |
| **Hardware, CAD & Tools:** | ESP32-S3, Arduino, LiDAR, MPU9250 / MPU6050 IMU, Encoder Motors, Barometer Sensors, NRF24L01, LoRa Module, ST7920 Display, Custom MOSFET ESC, Fusion 360, 3D Printing, TinkerCad, Git, Linux (Ubuntu 22.04), KML |

## EDUCATION

**PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur** `Aug 2024 – Aug 2028`

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering

- **Coursework:** Robotics (Kinematics, D-H Parameters, Trajectory Planning), Control Systems, Embedded Systems, Sensors & Actuators.
- **Practical Labs:** PUMA Robot Kinematics, Trajectory-Following Autonomous Mobile Robots.

## LEADERSHIP & EXPERIENCE

**Coordinator — Electronics and Robotics Society (ERS)** `Aug 2024 – Present`

PDPM IIITDM Jabalpur

- Led technical program for 200+ members; organized 10+ hands-on robotics workshops per semester on ROS2 and embedded systems.
- Mentored 30+ junior students in ROS2, SLAM, and autonomous systems; coordinated team participation for e-Yantra at IIT Bombay.

**Core Committee Member — Aero Fabrication Club** `Aug 2024 – Present`

PDPM IIITDM Jabalpur

- Designed and fabricated UAV hardware (frame assembly, ESC wiring, Pixhawk flight controller integration) for national competitions.

## FEATURED ROBOTICS PROJECTS

### [e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025–26) 🔗 [GitHub]] — Aug 2025 – May 2026

`ROS2 Humble · LiDAR · IMU · OpenCV · ArUco · MoveIt2 · UR5 Arm · Visual Servoing`

- Engineered a fully autonomous agricultural robot for crop monitoring and fruit harvesting; achieved **Top 16 / 400+ teams (Top 4% nationally)**.
- Built a multi-sensor perception pipeline (LiDAR + ultrasonic + IMU) enabling collision-free lane navigation across field rows.
- Developed ArUco marker-based fruit classification and automated pick-and-place routines with a UR5 arm via MoveIt2 and visual servoing.

### [Disaster Surveillance Drone — NIDAR 2026 🔗 [GitHub]] — Jan 2026 – Jun 2026

`Python · MAVSDK · ArduPilot SITL · MAVLink · QGroundControl · Mission Planner · YOLOv8 · KML`

- Architected a polygon-based autonomous survey UAV covering **75+ acres in <20 minutes** at 25m altitude via optimized zigzag mission paths.
- Integrated YOLOv8 real-time human detection with Return-To-Launch (RTL) failsafe, geofencing, and pause-center-resume target lock-on.
- Fully automated mission lifecycle (upload → arm → takeoff → survey → RTL) via MAVSDK Python API, reducing operator workload to zero.

### [Autonomous Drone Survey & Target Detection — SAE Aerothon 2025 🔗 [GitHub]] — Jan 2025 – Jun 2025

`Python · ROS2 · DroneKit · Pixhawk · ArduPilot · MAVLink · Gazebo · YOLOv4-Tiny · PID Control`

- Assembled a ROS2 autonomous drone for survey and target detection; selected for national-level competition presentation.
- Implemented pinhole camera model pixel-to-GPS conversion enabling sub-meter target geolocation from aerial imagery.
- Engineered a visual PID centering loop for precise autonomous target alignment and lock-on during hover operations.

### [Differential Drive Robot — ROS2 Navigation & SLAM 🔗 [GitHub]] (★ 7 Stars) — Jan 2024 – Jun 2024

`C++ · Python · ROS2 Humble · Gazebo · Nav2 · SLAM Toolbox · AMCL · URDF/xacro · EKF`

- Modeled and simulated a 4-wheeled differential drive robot with URDF/xacro physics model in Gazebo environment.
- Achieved SLAM-based 2D occupancy mapping using SLAM Toolbox and AMCL pose tracking for GPS-denied indoor localization.
- Deployed fully autonomous path planning, obstacle avoidance, and goal navigation using Nav2 stack with EKF sensor fusion.

### [ESP32-S3 Quadcopter Drone 🔗 [YouTube]] — Jun 2024 – Dec 2024

`ESP32-S3 · MPU6050 IMU · Barometer · Custom MOSFET ESC · Embedded C · TinkerCad`

- Fabricated a quadcopter drone from scratch with IMU stabilization and barometer altitude sensing.
- Designed a custom MOSFET-based Electronic Speed Controller (ESC) and flashed custom flight control firmware in Embedded C.

### [Modular Robotic Arm Platform 🔗 [GitHub]] — Aug 2024 – Present

`Servo Motors · Arduino · Kinematics · Trajectory Planning · 3D CAD`

- Developed a 5-DOF modular robotic arm with inverse kinematics for automated pick-and-place operations.

## KEY ACHIEVEMENTS & CERTIFICATIONS

🏆 **e-Yantra Robotics (IIT Bombay):** Top 16 / 400+ Teams (Certificate)

🤖 **DD Robocon 2025:** Stage-2 Qualification (Certificate)

✈️ **SAE Aerothon 2025:** Phase 2 Certificate

🚁 **NIDAR 2026:** National UAV Challenge (Certificate)